

Cyber Watchdog: Algorithm & Architecture Structure

Multi-Timescale Dynamical Systems (CRT Model) & Logistic Regression

Official Technical Documentation of the Risk Inference Engine

1. Architectural Strategy: The 5-Layer Digital Twin

The Cyber Watchdog implements a layered architecture to transform raw telemetry into predictive security insights. It models the human user as a **Dynamical System** where internal states (Stress, Energy, Habits) evolve over time driven by external events and biological constraints.

Layer 1: Multimodal Observation Space (X_t)

Every 10-second interval, the system captures a vector X_t consisting of 12 primary signals. These signals serve as the raw inputs for the state inference engine.

Sensor Family	Signal Parameter	Description & Impact
Work Context	1. Keystrokes	Total count of keys pressed. Indicates active workload.
	2. Mouse Entropy	Complexity of movement. High entropy = intense activity.
	3. Typing Errors	Backspaces/Deletes. High rate = cognitive friction/exhaustion.
	4. Task Switching	Window changes. High frequency = context-switching stress.
Environment	5. Notifications	Audio system alerts. Represents external interruptions.
	6. Email Patterns	Incoming frequency & unknown senders. High volume = pressure.
Digital Habits	7. Password Habits	Manager use (Good) vs Manual (Risk). Affects Habit state H_t .
	8. Browser Exposure	HTTP sites/Webmail. Direct indicator of security hygiene.
	9. OS Update Behavior	Delayed patches. Represents neglect of system health.
Biomarkers	10. Sleep Deficit	Working during rest hours. Reduces Capacity baseline μ .
	11. Phishing Sims	Interaction with bait links. Spikes Adversarial state A_t .
	12. Vision Fatigue	Blinks/Slumping (AI Camera). Rapidly drains Capacity C_t .

Layer 2 & 3: The CRT (Capacity, Reserve, Threat) Model

The system maps X_t into four hidden latent states Z_t . These states evolve using **Multi-Timescale Dynamics**, ensuring that risk predictions have temporal momentum.

A. Slow Dynamics: Cognitive Capacity (C_t)

$$dC/dt = \alpha (\mu - C_t)$$

Logic: C_t is your "energy tank." It drifts toward baseline μ . If **Sleep Deficit** or **Vision Fatigue** are detected, μ drops. Recovery speed α is much slower than drain speed.

B. Fast Dynamics: Cognitive Demand (D_t)

$$dD/dt = f(\text{work}) - \beta (D_t - 0.1)$$

Logic: D_t is current stress. **f(work)** is a weighted sum of **Keystrokes, Switches, Errors, and Notifications**. β is the "cooling rate" that pull stress back to idle (0.1) over time.

C. Medium Dynamics: Habit State (H_t)

$$H_{t+1} = H_t + \eta * \text{Reward}$$

Logic: Security hygiene. **Good Password Habits** give a positive reward; **Insecure Browser Exposure** or **Delayed Updates** give a penalty. η determines how fast the user "learns" or "forgets" good habits.

D. Core Output: Cognitive Reserve Gap (CRG_t)

$$CRG_t = C_t - D_t$$

Logic: The most critical safety indicator. **Positive CRG** = spare brainpower. **Negative CRG** = Cognitive Debt (extremely high risk of error).

Layer 4: Risk Forecasting Engine (Logistic Regression)

To convert complex hidden states into a single, actionable percentage, we use **Logistic Regression** with calibrated weights.

Step 1: The Linear Logit (z)

$$z = \beta_1 * CRG_t + \beta_2 * H_t + \beta_3 * A_t + \text{bias}$$

Weight	Value	Reasoning
β_1 (Reserve)	-3.0	High reserve (spare capacity) reduces risk linearly.
β_2 (Habits)	-3.0	Good habits act as a protective "shield" reducing risk.
β_3 (Threat)	+6.0	Active phishing click is a catastrophic risk multiplier.
Bias	+0.5	Base risk level assuming standard conditions.

Step 2: The Sigmoid "Squash"

We use the Sigmoid function to transform the Z-score into a probability $P(M_t)$ in the range [0.0, 1.0].

$$P(\text{Mistake}) = 1 / (1 + e^{-z})$$

Layer 5: Digital Twin Simulation & Architecture

Layer 5 uses the underlying CRT model to run **Counterfactual Simulations**. It projects the trajectory of the user state 30 days into the future to identify burnout points before they happen.

Unified Architecture Linkage:

Layer	Function	Algorithm Component
L1: Telemetry	Data Collection	The 12-Signal Vector X_t
L2/3: Inference	Hidden State Modeling	CRT Dynamical Systems (Euler Method)
L4: Forecasting	Instant Risk Score	Logistic Regression + Sigmoid
L5: Simulation	Predictive Trajectory	Heuristic Burnout Projections